

chip voltage/current monitors and protectors. Graphene SymFETs, which have a voltage-controlled unique peak current can be used to build low-cost, high-sensitivity circuit protectors through supply voltage monitoring.

Recent work has developed a SymFET-based power supply protector.^{1197,1198} With only two SymFETs, the power supply protector can easily monitor the supply voltage to ensure that the supply voltage to the circuit-under-protection is within a predefined range. In the event of a fault injection, the decreased supply voltage will power down the circuit rather than injecting a single-bit fault,¹¹⁹⁸ and can thus protect the circuit from fault injection attacks. If one uses V_{out} as the power supply to a circuit under protection (e.g., an adder), due to the bell-shaped I-V characteristic of the SymFET, an intentional lowering of V_{DD} cuts off the power supply. Thus, the sum and carry-out of the full adder output is '0', and no delay related faults are induced. A similar CMOS power supply protector would require op-amps for voltage comparison. As a result of the voltage/current monitors developed thus far, voltage/current-based fault injections can be largely prevented. By inserting the protectors in the critical components of a given circuit design, the power supply to these components can be monitored and protected.¹¹⁹⁹ (SymFET-based Boolean logic is also possible.¹²¹¹)

Preventing differential power analysis (DPA)—As an advanced side-channel attack scheme, DPA employs analysis of statistic power consumption measurements from a crypto system to obtain secret keys. Since the introduction of DPA¹²¹³, there has been many efforts to develop low-cost and efficient countermeasures. Countermeasures are generally classified into two categories: 1) algorithm-level solutions and 2) hardware-level solutions.

Algorithm-level solutions—aim to design cryptographic algorithms that can withstand a certain amount of information leakage,¹²¹² e.g., frequently changing the keys to prevent the attacker from collecting enough power traces¹²¹³ or using masking bits during the internal stages to limit information leakage.¹²¹⁴

A more practical circuit-level method for preventing DPA attack leverages a *sense amplifier-based logic* (SABL) or *current mode logic* (CML) for cryptographic algorithm implementations.¹²¹⁵ A CML gate includes a tail current source, a current steering core and a differential load. A CML gate will switch the constant current through the differential network of input transistors, utilizing the reduced voltage swing on the two load devices as the output. Although CML is not widely used in mainstream circuit design, its unique features, namely low latency and stable power consumption, can be leveraged to serve as a countermeasure against a DPA attack.

The strength of the CML-based approach is the constant power consumption of differential logic which can counter power-based attacks as operation power is independent of processed data. The drawback with these (mostly CMOS-based) logic designs, is their large area and power consumption when compared to static single ended logic. When considering hardware for the IoT (where the systems can be severely power constrained), system designers are presented with a dilemma in which they need to choose either high security or low power consumption. Emerging transistor technologies could help mitigate risks of DPA attacks while maintaining low power consumption.

Recent work has implemented a standard cell library of TFET CML gates and conducted a detailed study of their performance, power and area with respect to CMOS equivalents.¹²¹⁶ Standard cells were used to implement and evaluate TFET-based CML on a 32-bit KATAN cipher (a light-weight block cipher). All KATAN ciphers share the same key schedule with the key size of 80 bits as well as the 254-round iteration with the same non-linear function units.¹²¹⁷

The two CML implementations consume less gate equivalents and area compared to the two static counterparts given that the majority of KATAN32 is made up by the D flip flops. The area of TFET CML KATAN32 is $1.441 \mu\text{m}^2$, which is about 60% less than the Static TFET KATAN32. The power consumption of TFET CML ($9.76 \mu\text{W}$) is slightly lower than static CMOS ($9.96 \mu\text{W}$). It also outperforms CMOS CML.

Moreover, the correlation coefficient of a TFET static KATAN32 reaches its highest when the correct keys are applied. By comparison, the correlation coefficient of TFET CML KATAN32 is much more scattered, and all four hypothetical keys are equally distributed. Thus, the TFET CML KATAN32 implementation can successfully counteract CPA. Because the power consumption is mainly determined by AND/XOR logic gates of two nonlinear functions—and the effect of CPA is maximized—the correlation coefficients for KATAN32 are higher on average than other block ciphers, e.g., CPA on S-box¹²¹⁸.

6. EMERGING MATERIALS INTEGRATION

6.1. INTRODUCTION AND SCOPE

6.1.1. CURRENT STATE OF TECHNOLOGY

The semiconductor industry was historically driven by a strong correlation between technology scaling and performance of most integrated circuits. The PC market required more complex and faster microprocessors that largely drove the development and scaling of transistors and memory. These devices required new materials and processes such as strained silicon, high-k gate dielectrics and metal gate electrodes that are now widely, and will continue, to be used in IC manufacturing. In the past decade, a completely new ecosystem has emerged. New system integrators, from mobile to data centers to the Internet of Everything, have appeared with new and complex technology requirements. These system integrators will have impact that includes microprocessors, but extends towards new applications including medicine, energy and the environment.

6.1.2. DRIVERS AND TECHNOLOGY TARGETS

As transistors and memory begin to run out of horizontal space and ICs continue to be limited by power, device technologies will enter a phase characterized by vertical integration and performance specifications driven towards reduction of power. New transistor, memory, interconnect, lithography materials and processes will be required to enable this new More Moore scaling paradigm. As conventional information processing and storage technology reaches its ultimate limits, entirely new non-CMOS logic and memory devices and even new, non-Von Neumann circuit architectures are potential Beyond CMOS solutions. Such solutions ideally can be integrated onto the Si-based platform to take advantage of the established processing infrastructure, as well as being able to include Si devices such as memories onto the same chip. However, while these technologies will likely be integrated on a Si-based platform, the vast majority of these Beyond CMOS technologies are based on entirely new materials and physics. Finally, new system integrators require materials that enable potentially trans-disciplinary advances in monolithically integrated complex functionality, i.e., functional scaling. Significant challenges must be overcome for these emerging materials to provide viable solutions for future integrated circuit technologies. To deliver these capabilities, enhanced Metrology will be needed to accelerate material evaluation, improvement and capabilities. The ultimate goal is to provide timely guidance on emerging material and process performance, cost, reliability, and sustainability options that will drive breakthrough advances in future manufacturing technology.

6.1.3. SCOPE

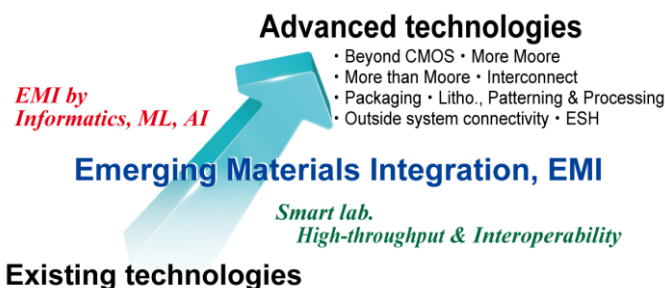


Figure EMII Emerging Material Integration Promotes the Advancement of Existing Technologies

The IRDS represents a strategic repositioning of the community's scope, needs, and set of emergent opportunities. In alignment with this new perspective, this edition of the emerging materials integration (EMI) sub-chapter represents a work in transition with a primary goal of aligning with the needs of related IRDS working groups. Much of the associated information in the detailed requirements and solutions tables comes from prior ERM chapters and input from current IRDS working groups, and will be updated in future editions. The chapter emphasizes strategic difficult challenges and/or enabling of novel, breakthrough and potentially disruptive opportunities for emerging material properties, synthetic methods, and metrology, organized in the following areas:

1. **Scaled technology materials needs for More Moore:** transistors, memory, interconnects, lithography, heterogeneous integration, assembly and packaging.
2. **Novel materials for Beyond CMOS:** emerging logic and information processing devices, emerging memory and storage devices, and novel computational paradigms and architectures.
3. **Potentially disruptive material opportunities for functional scaling and convergent applications:** Heterogeneous components, outside system connectivity, and high impact application areas such as energy, environment, agriculture, health, medical, etc.

For all areas, the advancement requires an intergration of emerging materials as illustrated in Figure EMI 1.

6.2. CHALLENGES

6.2.1. NEAR-TERM CHALLENGES

Table EMI1 *Near-term Difficult Challenges*

Near-Term Challenges: 2022–2028	Description
<i>Materials and processes that achieve performance and power scaling of lateral fin- and nanowire FETs (Si, SiGe, Ge, III-V).</i>	Integrated high k dielectrics with EOT <0.5nm and low leakage. Integrated contact structures that have ultralow contact resistivity. Achieving high hole mobility in III-V materials in FET structures. Achieving high electron mobility in Ge with low contact resistivity in FET structures. Processes for achieving low dislocations and anti-phase boundary generating interface between Ge/III-V channel materials and Si. Dopant placement and activation i.e., deterministic doping with desired number at precise location for V _{th} control and S/D formation in Si as well as alternate materials.
<i>Materials and processes that improve copper interconnect resistance and reliability</i>	Mitigate impact of size effects in interconnect structures. Patterning, cleaning, and filling at nano dimensions. Cu wiring barrier materials must prevent Cu diffusion into the adjacent dielectric but also must form a suitable, high quality interface with Cu to limit vacancy diffusion and achieve acceptable electromigration lifetimes. Reduction of the k value of inter-metal dielectrics.
<i>Materials and processes for continued scaling of DRAM/SRAM and embedded NVM</i>	Low temperature materials for high performance vertical transistor memory select structures. High-k, low leakage DRAM dielectrics. Processes for stacking of 3D flash.
<i>Materials and processes that improve AI and Quantum computing</i>	Integrate ultra-low power consumption AI devices for advanced edge computing. Development of process technology for introducing various new-principle computing materials, including liquids and biomaterials, into existing electronics. Elucidation of device physics at low temperatures to realize low-temperature electronics.
<i>Materials and processes that extend lithography to sub-10 nm dimensions with reproducible properties</i>	Novel resists to extend 193 nm lithography and support EUV lithography. Directed self-assembly (DSA) with materials such as block-copolymers to potentially extend lithography though pattern rectification and pattern density multiplication.
<i>Materials for heterogeneous integration of multi-chip, multi-function packages.</i>	Materials to modify polymer properties to enable increased product reliability. Novel electrical attaching materials to allow lower assembly temperatures and improved product reliability. Simultaneously achieve package polymer CTE, modulus, electrical and thermal properties, with moisture and ion diffusion barriers. Nanosolders compatible with <200C assembly, multiple reflows, high strength, and high electromigration resistance. Nanoinks that can be printed as die attach adhesives with required electrical, mechanical, thermal, and reliability properties.

6.2.2. LONG-TERM CHALLENGES

Table EMI2 *Long-term Difficult Challenges*

Long-term Challenges: 2029–2037	Description
<i>Materials and processes that achieve 3D monolithic and vertical integration of high mobility and steep subthreshold transistors</i>	Processes for sequential 3D vertical integration of transistors. Methods to lower the synthesis temperature of vertical semiconductor nanowires. Methods to dope and contact vertical semiconductor nanowire transistors. Lithography-free and low-temperature methods to achieve gate stack on vertical transistors.
<i>Materials and processes that replace copper interconnects with improved reliability and electromagnetic performance at the nanoscale</i>	Synthesis or assembly of CNTs in predefined locations and directions with controlled diameters, chirality and site-density. Carbon and collective excitations. Novel interlayer dielectrics: Metal Organic Framework (MOF) and Carbon Organic Framework (COF). Metals with less size effects such as silicides.
<i>Materials and processes for charge-based and non-charge-based beyond CMOS logic that replaces or extends CMOS</i>	Achieving a bandgap and full interfaces control in graphene in FET structures and alternative FETs (TFETs etc). Synthesis of CNTs with tight distribution of bandgap and mobility. Complex metal oxides with low defect density. High mobility transition metal dichalcogenides with low defect density and low resistance ohmic contacts. Spin materials: characterization of spin, magnetic and electrical properties and correlation to nanostructure. Topological materials: large bandgaps much greater than kT at room temperature, ability to modulate bandgap efficiently with electric field. BiSFET heterostructures: achieving exciton condensation at room temperature.
<i>Materials and processes for emerging memory and select devices to replace DRAM/NVM.</i>	Multiferroic with Curie temperature >400K and high remnant magnetization to >400K. Ferromagnetic semiconductor with Curie temperature >400K. Complex Oxides: Control of oxygen vacancy formation at metal interfaces and interactions of electrodes with oxygen and vacancies. Switching mechanism of atomic switch: Improvements in switching speed, cyclic

<i>Long-term Challenges: 2029–2037</i>	<i>Description</i>
	endurance, uniformity of the switching bias voltage and resistances both for the on-state and the off-state.
<i>Materials and processes that enable monolithically 3D integrated complex functionality including thermal and yield challenges</i>	Integration on CMOS Platforms. Integration with flexible electronics. Biocompatible functional materials. Leveraging convergent materials expertise in adjacent sectors, including More than Moore functionalities (photonics, optics/metamaterials, outside connectivity, energy transfer/storage, power circuits).
<i>Green and sustainable fabrication</i>	To realize an integration process that is environmentally friendly and economically viable. Utilizing AI, ML, and informatics to develop production processes that can introduce green new materials.

6.3. TECHNOLOGY REQUIREMENTS AND POTENTIAL SOLUTIONS

6.3.1. SUMMARY

The IRDS seeks a framework for managing the convergence of scaled information processing and storage, i.e., More Moore (MM) and Beyond CMOS (BC), with the next emerging era of monolithically integrated systems that achieve enhanced overall functional density. The trend towards the convergence of monolithically integrated functional diversification with miniaturization manifests as increasing complexity in the road-mapping process. The IRDS reflects this growing complexity, with an increasing number of projected roadmap parameters and requirements associated with new functionalities. While EMI continues to support the evolutionary, and semiconductor centric needs of the traditional semiconductor community, emerging architectures could benefit from new device functionality, which may require new materials and new physical mechanisms. New waves of emerging materials technologies may represent potentially disruptive opportunities.

Candidate EMI materials and processes exhibit unique and useful properties that may require atomic level structural, interface, defect and compositional control. In some cases, current synthetic or manufacturing technologies are not yet capable of producing such materials with the required level of control. The difficulties could be due to: 1) The inability of a research environment to produce materials with the required level of control that would express the desired properties; or 2) scaling up the synthetic and fabrication processes to satisfy commercial manufacturing requirements. In some cases, current materials growth processes effect unacceptable levels of defect formation, which drive the need for new and more robust fabrication methods. In other cases, synthetic methods exist for producing high quality materials, but these processes cannot be scaled to the higher growth rates, yields, or purity needed for insertion into viable commercial applications. While these materials may provide proof of concept and suggest a potential solution, new cost-effective fabrication technologies may be required to warrant a candidate material's insertion into high volume manufacturing. It will be an extremely important issue to be able to quickly verify the integration process suitability of emerging materials in large-scale semiconductor factories. For this reason, it is important to establish standardized indices for evaluating the characteristics of new materials and new devices. We should also consider using the international standards being conducted at TC113 (Technical committee No.113, Nanotechnology for electrotechnical products and systems) of the International Electrotechnical Commission (IEC).

6.3.2. SCALED TECHNOLOGY MATERIALS FOR MORE MOORE

As described in the More Moore chapter, after 2027 there is no headroom for 2D geometry scaling and 3D VLSI integration of circuits and systems using sequential/stacked integration approaches will likely begin. Whether one is considering 2D geometry scaling or 3D integration, there are numerous materials challenges to achieving increasing device density and integrated performance. The following outlines key materials challenges for transistor scaling and integration, lithography, interconnects, heterogenous integration, assembly and packaging, and outside system connectivity.

6.3.2.1. MATERIALS FOR TRANSISTOR SCALING AND INTEGRATION

Continued increases in transistor device density require a variety of new materials and processes including new channels (Ge, III-V), improved doping techniques, gate stacks and contacting structures. Table EMI3 provides a set of materials and processes priorities for transistor scaling and integration.

Table EMI3 Materials for Transistor Scaling and Integration

6.3.2.2. MATERIALS FOR LITHOGRAPHY AND PATTERNING

The future of scaled technologies depends upon emerging patterning materials (resist or self-assembled) to enable extensible lithographic capabilities. New resist materials must concurrently exhibit higher resolution, higher sensitivity, reduced line edge

roughness, and sufficient etch resistance to enable robust pattern transfer. 193nm and EUV extension materials are being developed, which can improve LWR, pattern shrink materials, and topcoats for EUV to ameliorate issues with out-of-band optical flare and outgassing. Evolutionary approaches for enhancing positive, negative, and chemically amplified families of resists will continue to be evaluated. Leading process approaches to pitch division include multiple patterning (MP) and spacer patterning (SP) as options for extending 193 nm immersion lithography. Alternate technologies are utilizing patterning materials to create guide patterns for directed self-assembly, which can include resists to form chemoepitaxy and graphoepitaxy guides, or directly patternable brushes and SAMs. Directed self-assembly (DSA) with block-copolymers or polymer pairs has made significant progress in characterizing sources of defect formation and in applications such as contact rectification, fin patterning, and pattern density multiplication. Table EMI4 provides a set of materials and processes priorities for lithography and patterning.

Table EMI4 Materials for Lithography and Patterning

6.3.2.3. INTERCONNECT MATERIALS

Key challenges for continued increased performance of future integrated circuit interconnects consist of maintaining reductions of RC time constants for delivery of signals and power with high reliability. For copper interconnects, the sidewall copper barrier thickness must continue to be reduced, which is a significant challenge. For post copper interconnect scaling, novel interconnects, such as carbon nanotubes, are being explored. Several elemental metals have been studied by simulation and experiments to identify potential candidates for post-Cu interconnect materials at highly scaled dimension.¹²¹⁹ The interconnect material space becomes significantly larger when alloys and compounds are considered, and machine learning assisted data analysis has been employed to filter this vast material space. Also, lower dielectric constant (κ for both intra and inter level dielectric are needed; however, each of these emerging families of materials must overcome significant challenges for them to warrant adoption. Airgap, another approach to reducing the effective κ , places additional requirements on barrier layers or novel interconnects. Table EMI5 provides a set of materials and processes priorities for interconnects.

Table EMI5 Interconnect Materials

6.3.2.4. HETEROGENEOUS INTEGRATION, ASSEMBLY AND PACKAGING MATERIALS

The EMI and Heterogeneous Integration teams are in the process of prioritizing key heterogeneous integration and assembly & packaging EMI challenges, which include:

- New engineered materials: substrate, mold, underfill, wafer bond alloys, solder alloys
- Conductors: Nanomaterials (CNT, graphene, NWs), metals (Cu, Al, W, Ag, etc.), composites
- Dielectrics: Oxides, polymers, porous materials, composites
- Semiconductors: Elemental (Si, Ge), Compounds (SiC, III-V, II-VI, tertiary), polymers
- Critical factors: Cost, CTE differential, thermal conductivity, fracture toughness, modulus, processing temperature, interfacial adhesion, operating temperature, and breakdown field strength

Table EMI6 provides a set of heterogeneous integration and assembly and packaging priorities for EMI.

Table EMI6 Heterogeneous Integration, Assembly and Packaging Materials

6.3.2.5. MATERIALS CHALLENGES FOR OUTSIDE SYSTEM CONNECTIVITY

Table EMI7 provides a set of top Outside System Connectivity material priorities for EMI.

Table EMI7 Emerging Research Materials Needs for Outside System Connectivity

6.3.3. EMERGING MATERIALS FOR MEMORY, BEYOND CMOS LOGIC AND COMPUTING

Beyond 2030, MOSFET scaling will likely become ineffective and/or very costly. As described in this chapter, completely new, non-CMOS types of memory, logic devices and maybe even new circuit architectures are potential solutions. Such solutions ideally can be integrated onto the Si-based platform to take advantage of the established processing infrastructure, as well as being able to include Si devices such as memories onto the same chip. The following outlines key materials challenges for emerging materials for memory, beyond CMOS logic and alternative information processing.

6.3.3.1. EMERGING MATERIALS FOR MEMORY

Emerging memory devices includes capacitive memories (Fe FET), and resistive memories including ferroelectric devices, resistance change devices, devices based on Mott transitions and novel magnetic memories. Another key requirement for memory technology is the development of corresponding select devices that access only the selected memory cell of interest without perturbing non-selected cells. Table EMI8 provides a set of materials and associated challenges for emerging memory materials, and Table EMI9 provides materials and associated challenges for memory select.

Table EMI8 Emerging Materials for Memory

Table EMI9 Emerging Materials for Memory Select

6.3.3.2. EMERGING MATERIALS FOR ADVANCED AND BEYOND-CMOS LOGIC DEVICES

There are generally two classes of devices/materials for advanced and beyond-CMOS logic devices. The first are those that do not involve spin or magnetism such as ferroelectric FETs, nanoelectromechanical (NEM) switches, topological FETs, and transistors based on collective electron phenomena such as Mott effect or exciton condensation (BiSFET). The second are those based on spin and magnetism that each uses a variety of materials. Negative-capacitance topological quantum FET was proposed to achieve extremely low switching voltages and energies.¹²²⁰ Table EMI10 contains materials and associated challenges for the non-spin devices. Table EMI11 maps various spin device concepts to associated material types and Table EMI12 describes the requirements of these materials.

Table EMI10 Emerging Materials for Advanced and Beyond-CMOS Logic Devices

Table EMI11 Spin Devices versus Materials

Table EMI12 Spin Material Requirements and Properties

6.3.3.3. EMERGING MATERIALS FOR NOVEL COMPUTING

Emerging materials spur developments of novel computing, such as neuromorphic computing, reinforcement learning, topological quantum computing, and reversible computing, and probabilistic computing. Since the performance of the computing is considered to be dependent on the intrinsic properties of the emerging material, the material research will further boost the performance, such as the energy efficiency^{1221,1222,1223,1224,1225,1226}. Two-dimensional materials are also expected to play an important role in memristors for neuromorphic computing.^{1227, 1228}

6.3.3.4. NANOPARTICLE-BASED COMPUTING

The implementation of complex computation in particle systems, especially in nanoparticles, remains challenging. However, nanoparticle-based computing has a wide range of potential applications that benefit from algorithmically controlling their unique and potentially useful intrinsic properties (photonic, plasmonic, catalytic, photothermal, optoelectronic, electrical, magnetic, material properties, etc.) without human intervention^{1229, 1230,1231,1232}.

6.3.4. METROLOGY NEEDS AND CHALLENGES FOR EMERGING RESEARCH MATERIALS

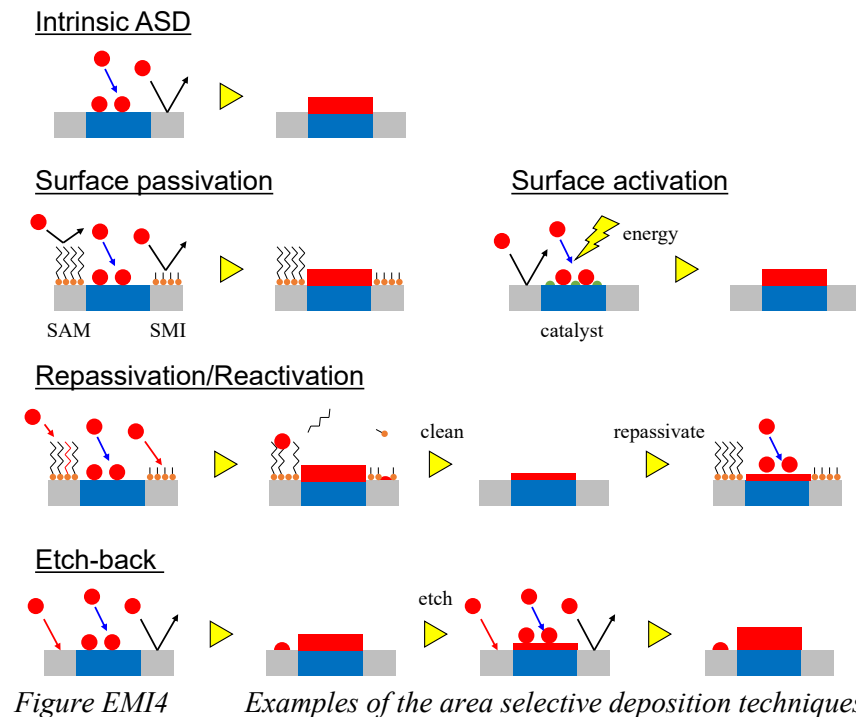
Metrology is needed to characterize composition, properties, and understand structure of emerging research materials, at nanometer dimensions and below. The difficult EMI metrology challenges would be those associated with the introduction of directed self-assembly (DSA), such as evaluating critical material properties, size and location of features, registration, and defects. Also needed are non-destructive methods for characterizing embedded materials and interfaces defects in nano-scale devices,¹²³³ as well as platforms that enable simultaneous measurement of complex nanoscopic properties. Table EMI13 summarizes the current set of continuing and prioritized metrology related EMI challenges and needs.

Table EMI13 Metrology Needs and Challenges for Emerging Research Materials

6.3.5. NEEDS AND CHALLENGES OF EMERGING RESEARCH MATERIALS FOR ESHS

Research and development of Beyond CMOS devices will be stout under limitation of Green Materials. Therefore, the close liaison with Environment, Safety, Health and Sustainability (ESHS) will continue to be an important issue.

and activation techniques decay gradually during the deposition process, appropriate repassivation and reactivation are usually required. If small amounts of film are formed in undesirable locations, they can be removed by the etch-back process. In any combination of techniques, the area selective deposition property of the precursor compounds will play an important role in the success of the deposition process. ASD processes are classified into several groups according to the combination of substrates (metals/ dielectrics/ semiconductors) to activate and passivate the deposition: dielectric on dielectric (DoD),¹²⁵¹ metal on metal (MoM),¹²⁵² metal on dielectric (MoD),¹²⁵³ dielectric on semiconductor (DoS), etc. The ASD process will help reduce takt time and waste through efficient formation of, for example, contacts, hard masks, and via-fills.



6.5. CONCLUSIONS AND RECOMMENDATIONS

The IRDS represents a strategic repositioning of the community's scope, needs, and set of emergent opportunities. In alignment with this new perspective, this edition of the EMI chapter represents a work in transition that has aligned difficult challenges with the needs of related IRDS working groups. Much of the information in the detailed tables comes from prior ERM chapters and current IRDS working groups. Future editions of EMI will provide additional detailed descriptions and continue to adapt its scope to engage with a new set of EMIs, many of which will be identified by the IRDS working groups.

7. ASSESSMENT

7.1. INTRODUCTION

It is important to evaluate beyond-CMOS devices considered in this chapter against current CMOS technologies. Two methods of assessments have been reported: a "survey-based assessment" conducted previously by the ITRS ERD working group and a "quantitative device benchmarking" conducted by the Nanoelectronics Research Initiative (NRI).

Up to the 2013 ERD Chapter, a survey-based critical review was conducted based on eight criteria to compare emerging devices against CMOS. Spider chart has been used to visualize the perceived potential of these technology entries. However, the limited number of survey results sometimes raises questions of the accuracy of this survey. The most recent "survey-based assessment" was conducted in the 2014 ERD Emerging Memory and Logic Device Assessment Workshops (Albuquerque, NM). The survey collects voting on emerging technologies evaluated in the workshops in the categories of the "most promising" and the "most need of resources" to assess the potential of these technology entries perceived by ERD experts. A summary of previous survey-based assessments is included in section 7.3 "Archive".

In the "NRI benchmarking", each emerging device is evaluated by its operation in conventional Boolean Logic circuits, e.g., a unity gain inverter, a 2-input NAND gate, and a 32-bit shift register. Metrics evaluated include speed, areal footprint, power dissipation, etc. Each parameter is compared with the performance projected for high-performance and low-power 5 nm CMOS.

Most of the proposed beyond-CMOS devices are very different from their CMOS counterparts, and often pass computational state variables (or tokens) other than charge. Alternative state variables include collective or single spins, excitons, plasmons, photons, magnetic domains, polarization, etc. With the multiplicity of programs characterizing the physics of proposed new structures, it is necessary to find ways to benchmark the technologies effectively. This requires a combination of existing metrics used for CMOS and new metrics which take into consideration the idiosyncrasies of the new device behavior. Even more challenging is to extend this process to consider new circuits and architectures beyond the Boolean logic used by CMOS today, which may enable these devices to complete transactions more effectively.

7.1.1. ARCHITECTURAL REQUIREMENTS FOR A COMPETITIVE LOGIC DEVICE

The circuit designers and architects depend on the logic switch to exhibit specific desired characteristics in order to successfully realize a wide range of applications. These characteristics,¹²⁵⁴ which have since been supplemented in the literature, include:

- Inversion and flexibility (can form an infinite number of logic functions)
- Isolation (output does not affect input)
- Logic gain (output may drive more than one following gate and provides a high I_{on}/I_{off} ratio)
- Logical completeness (the device is capable of realizing any arbitrary logic function)
- Self-restoring / stable (signal quality restored in each gate)
- Low cost manufacturability (acceptable process tolerance)
- Reliability (aging, wear-out, radiation immunity)
- Performance (transaction throughput improvement)
- Span of control (measures number of devices that may be reached within a characteristic delay of the switch¹²⁵⁵)

Devices with intrinsic properties supporting the above features will be adopted more readily by the industry. Moreover, devices which enable architectures that address emerging concerns such as computational efficiency, complexity management, self-organized reliability and serviceability, and intrinsic cyber-security¹²⁵⁶ are particularly valuable.

7.2. NRI BEYOND-CMOS BENCHMARKING

The Nanoelectronics Research Initiative (<https://www.src.org/program/nri/>) has quantitatively evaluated quite a few beyond-CMOS technologies.^{1257, 1258, 1259} Several NRI devices have been described in detail in the Logic and Emerging Information Processing Device Section. While beyond-CMOS device benchmarking is still very much a work in progress – and no concrete decisions have been made on which devices should be chosen or eliminated as candidates to significantly extend or augment CMOS – this section summarizes some of the data and insights gained from these studies. Further benchmarks may alter some of the conclusions here and the outlook on some of these devices, but the overall message on the challenges of beyond-CMOS devices remains valid.

7.2.1. QUANTITATIVE RESULTS

NRI benchmarking analyzes the potential of major emerging switches using a variety of information tokens and communication transport mechanisms. Specifically, the projected effectiveness of these devices used in a number of logic gate configurations was evaluated and normalized to CMOS at the 5nm generation (projection). The initial work has focused on “standard” Boolean logic architecture, since the CMOS equivalent is readily available for comparison. It should be noted that the majority of devices are evaluated via simulations since many of them have not yet been built, so it should be considered only a “snapshot in time” of the potential of any given device. Data on all of them are still evolving.

At a high level, the data from these studies corroborates qualitative insights from earlier works, suggesting that many new logic switch structures may have some advantages over CMOS in terms of power or energy, but they are also inferior to CMOS in delay. This is perhaps not surprising; the primary goal for nanoelectronics and NRI is to find a lower power device¹²⁶⁰ since power density is a primary concern for future CMOS scaling. The power-speed tradeoffs commonly observed in CMOS also extend to beyond-CMOS devices. It is important to understand the impact of transport delay for the different information tokens these devices employ. Communication with many non-charge tokens can be significantly slower than moving charge, although this may be balanced in some cases with lower energy for transport. The combination of the new balance between switch speed, switch area, and interconnect speed can lead to advantages in the span of control for a given technology. For some of the technologies (e.g., nanomagnetic logic), there is no strong distinction between the switch and the interconnect, indicating the need for novel architecture to exploit unique attributes of these technologies.

A simplified 32-bit arithmetic logic unit (ALU) was built from these devices to evaluate their performance, as shown in Figure BC7.1(a)¹²⁶¹. While tunneling devices show limited advantages over CMOS in terms of energy-delay product, most beyond-

CMOS devices are inferior to CMOS in energy and/or delay. For example, spintronic devices tend to be slower than CMOS and also show no energy advantage.

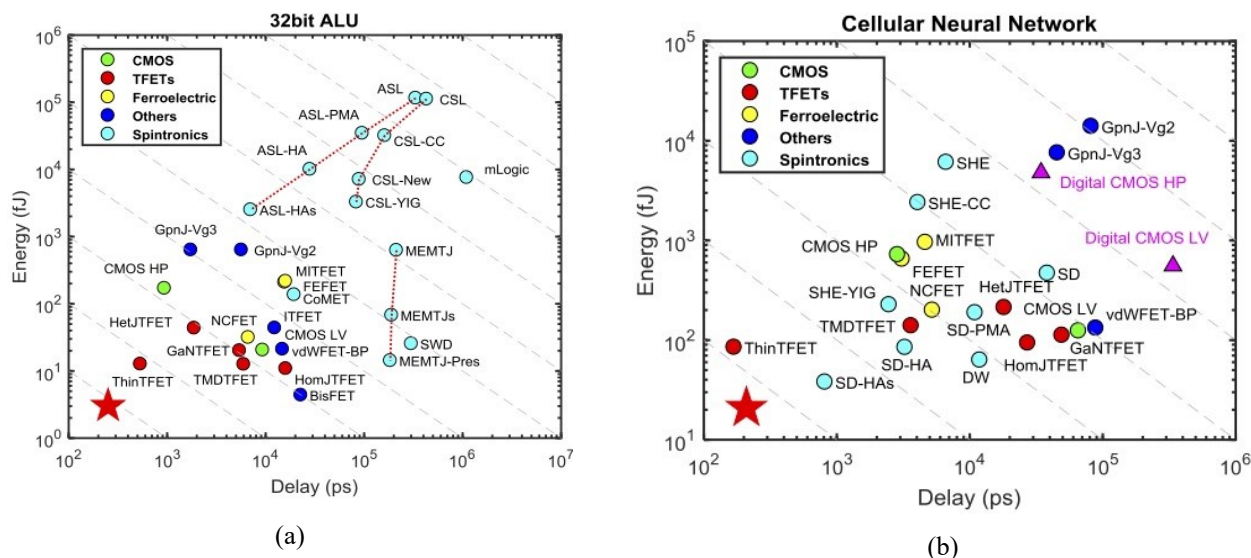


Figure BC7.1 (a) Energy versus Delay of a 32-bit ALU for a Variety of Charge- and Spin-based Devices; (b) Energy versus Delay per Memory Association Operation Using Cellular Neural Network (CNN) for a Variety of Charge- and Spin-based Devices¹²⁶¹

At the architecture level, the ability to speculate on how these devices will perform is still in its infancy. While the ultimate goal is to compare at a very high level—e.g., how many MIPS can be produced for 100 mW on 1 mm²?—the current work must extrapolate from only very primitive gate structures. One initial attempt to start this process has been to look at the relative “logical effort”¹²⁶² for these technologies, a figure of merit that ties fundamental technology to a resulting logic transaction. Several of the devices appear to offer advantage over CMOS in logical effort, particular for more complex functions, which increases the urgency of doing more joint device-architecture co-design for these emerging technologies.

The direction of device-architecture co-optimization has driven NRI benchmarking to explore non-Boolean applications of beyond-CMOS devices. Cellular neural network (CNN) has been utilized as a benchmarking model that has been implemented with various novel devices.¹²⁶¹ The energy and delay of CNN based on beyond-CMOS devices are compared with CMOS-based CNN in Figure BC7.1(b). Tunneling devices have significant performance improvement because of their steep subthreshold slopes and large driving current at ultra-low supply voltage. Interestingly, spintronic devices are much closer to the preferred corner in CNN implementation in comparison with 32-bit ALU. This is because some characteristics of spintronic devices (e.g., spin diffusion, domain wall motion) may mimic the functionality of a neuron (e.g., integration) more naturally in a single device.

7.2.2. OBSERVATIONS

A number of common themes have emerged from these benchmarking studies and in the observations made during recent studies of beyond-CMOS devices.

1. The low voltage energy-delay tradeoff conundrum will continue to be a challenge for all devices. Getting to low voltage must remain a priority for achieving low power, but new approaches to achieve sufficient throughput with “slow” devices must be developed.
2. Most of the architectures that have been considered to date in the context of new devices utilize binary logic to implement von Neumann computing structures. In this area, CMOS implementations are difficult to supplant because they are very competitive across the spectrum of energy, delay and area – not surprising since these architectures have evolved over several decades to exploit the properties of CMOS most effectively. Novel electron-based devices—which can include devices that take advantage of collective and non-equilibrium effects—appear to be the best candidates as a drop-in replacement for CMOS for binary logic applications.
3. As the behavior of other emerging research devices becomes better understood, work on novel architectures that leverage these features will be increasingly important. A device that may not be competitive at doing a simple NAND function may have advantages in doing a complex adder or multiplier instead. Understanding the right building blocks for each